

Bachelor of Technology (Computer Science & Engineering)

CSD493: Project-1, Monsoon 2026

Project Proposal Form

Date: 15th June 2026

Project Title: Behavioral-Diff Detection of Malicious Browser Extension Updates

Area of Specialization: Cybersecurity

Names and SNU IDs of Group Members (in capital letters):

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Project Proposal Description:

MOTIVATION:

Browser extensions update silently and automatically across every major browser. A trusted extension can become malicious overnight - the December 2024 Cyberhaven supply-chain attack pushed a credential-stealing payload to 400,000 active users within hours, entirely undetected, and the same campaign compromised dozens of other Chromium extensions. The browser's trust contract is granted once at install time and silently inherited by every later update, with no re-consent from the user and no re-review by the store. Static code review and permission analysis fail against obfuscated payloads and behaviours that only surface at runtime under specific browsing conditions. There is currently no open, systematic, *version-aware* tool that detects malicious changes between extension versions through dynamic behavioural comparison - and none that does so across multiple browser families.

PROBLEM STATEMENT:

This project builds a sandboxed behavioural-diff framework that flags malicious browser-extension updates by comparing runtime traces across consecutive versions, across multiple browser families: Chromium-based browsers (Chrome, Edge, Brave) and Firefox. For each browser, the system executes both version N and version $N+1$ inside an isolated Docker container under identical, record/replayed Playwright-automated browsing tasks (targeting Manifest V3), recording outbound network calls, DOM field access, extension-API usage, and storage/cookie access for each version. To surface dormant logic, it actively triggers condition-gated payloads through synthetic DOM interaction (dummy credential/login forms over Tranco-ranked

sites, financial login pages, and webmail portals). A weighted feature-difference vector $\Delta = f(T_{N+1}) - f(T_N)$ is computed between the two traces and scored both by a transparent weighted baseline and by a Random Forest classifier, trained primarily on a synthetic weaponisation corpus and validated on confirmed malicious update pairs from the 2024–25 supply-chain incidents. The output is an explainable per-update diff report that specifies exactly which new behaviours triggered the alert - giving security teams actionable intelligence rather than opaque risk scores. The system is evaluated on detection rate, false-positive rate, cross-browser consistency, evasion resistance, and per-update latency overhead.

Signature of Group Member(s), Date

Signature of Project Advisor, Date